POWERLABS

W90 INVERTER

User Manual

Rev. 0.6 - December 2023

INDEX

1. INTRODUCTION

The current document provides the needed information for the correct use and operation of the W90 inverters manufactured by EPowerlabs. It also specifies the requirements and the use limitations of the device.

This version replaces all previous versions. EPowerlabs has made every effort to ensure this document is complete and accurate. In accordance with our policy of continuing product improvement, all data in this document is subject to change or correction without prior notice.

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2. SAFETY AND WARNING

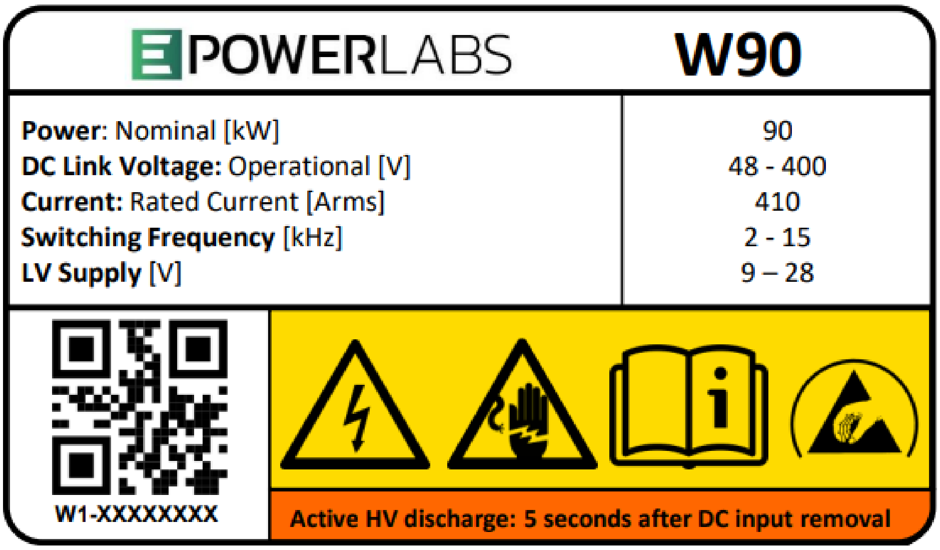
This device is able to conduct high voltages and currents, which means that under improper manipulation it can be the source of injury and/or damage to people, animals, or objects. To avoid this, the following instructions and recommendations are provided.

Note that the manipulation of the inverter should be carried out by a qualified professional in full compliance with the established legislation and technical regulations as well as the safety regulations governing electrical equipment. Please note that the present literature supplements and does not replace any established legislation, technical regulations or safety regulations.

- Although the device is designed to support mechanical stress and vibrations, the device is still fragile, and can get damaged if exposed to strokes of falls.
- The inverter can reach high temperatures. Allow sufficient time for the inverter to cool when having to touch the housing or any of the auxiliary elements.
- The voltages on the inverter may surpass 500V, and currents may surpass 500A. Staff who operates with it should be conscious of the electrical risks and be able to act in order to avoid any kind of danger.
- It is not recommended to open the housing cover without the warrant of the manufacturer. In case the cover is removed, the operator should be aware about the ESD risk that could damage the inverter and Inverter warranty is voided.
- There shall be a direct and strong grounding connection between motor housing and inverter housing.
- In case of cooling leakage, stop the device immediately and contact EPowerlabs for assistance.
- Do not connect or disconnect HV connections before ensuring that there is no active voltage in the inverter.

3. PRODUCT IDENTIFICATION LABEL

Each inverter features an identification label with a unique serial number, main operational ratings of the inverter and warning signs the user needs to bear in mind.



4. MAIN FEATURES

The W90 inverters is a high-power density traction inverter which delivers up to 90 kW power in a small package for 400V electric machines. It can be used for test bench operation, development activities or mounted to vehicle.

- Sixth gen. IGBT powerpack.
- Control board triple core @300Mhz (ASIL-D capable)
- Redundancy functions for safety-critical applications
- Real time parameter calibration and monitoring
- PC based set up and programming tools available.
- Application software enables sensors auto-calibration functions.
- Customizable control algorithms for each application
- Position sensor types: Resolver, Absolut Encoder, Incremental Encoder, XMR, BiSS.
- Current sensor: hall
- Temperature sensor: NTC 10k, PT1000. Can be calibrated for other sensors upon request.
- Id/Iq Current, Torque and Speed control.
- Self-protection features
- 2x CAN communication channels (Communication and DeveLinkSTUDIO calibration tool).
- CAN database available (.dbc)
- It can control any type of 3 phase PMSM, ACIM or BLDC motors.

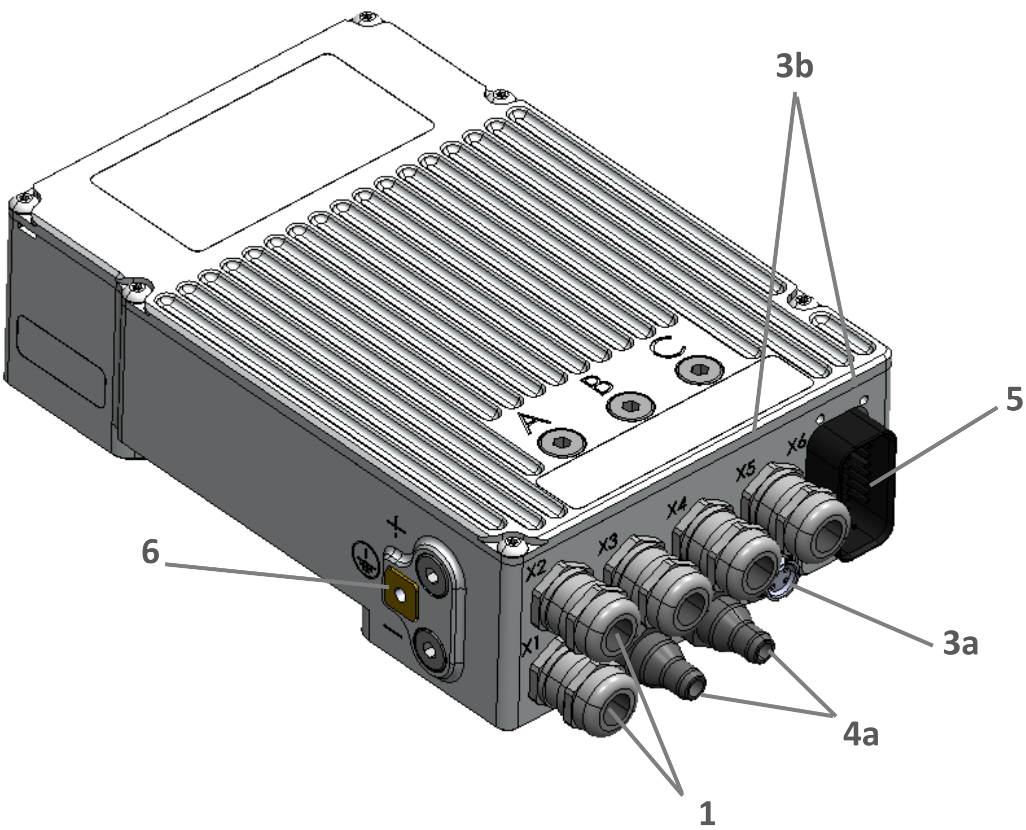
5. RATINGS AND DIMENSIONS

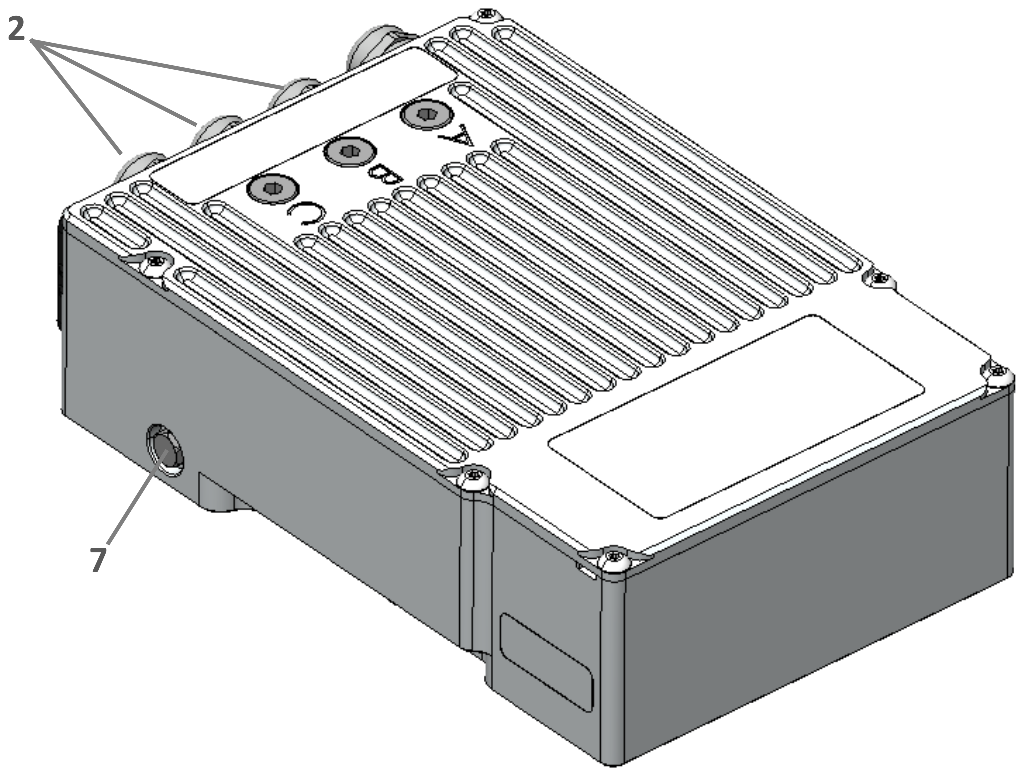
5.1. Electrical General Data

Description	W90
Nominal DC link voltage	48 – 400 [VDC]
Maximum passive DC link voltage	500 [VDC]
Maximum DC link voltage for operation	440 [VDC]
Rated Current	410 [Arms]
Continuous Power @400V	90 [kW] [1]
Converter Switching Frequency	2-15 [kHz]
LV Supply	9-28 [VDC]
Supply current at 12V pulsing 12kHz	1200 [mA]
Maximum electrical frequency for the entire operational temperature range	1600 [Hz]
Input DC Link capacitance	380-420 [uF]

1) Flow: 8l/min, F_{swi}: 10kHz, $T_{cooling}$: 25ºC, T_{air}: 20ºC

5.2. Connector Overview

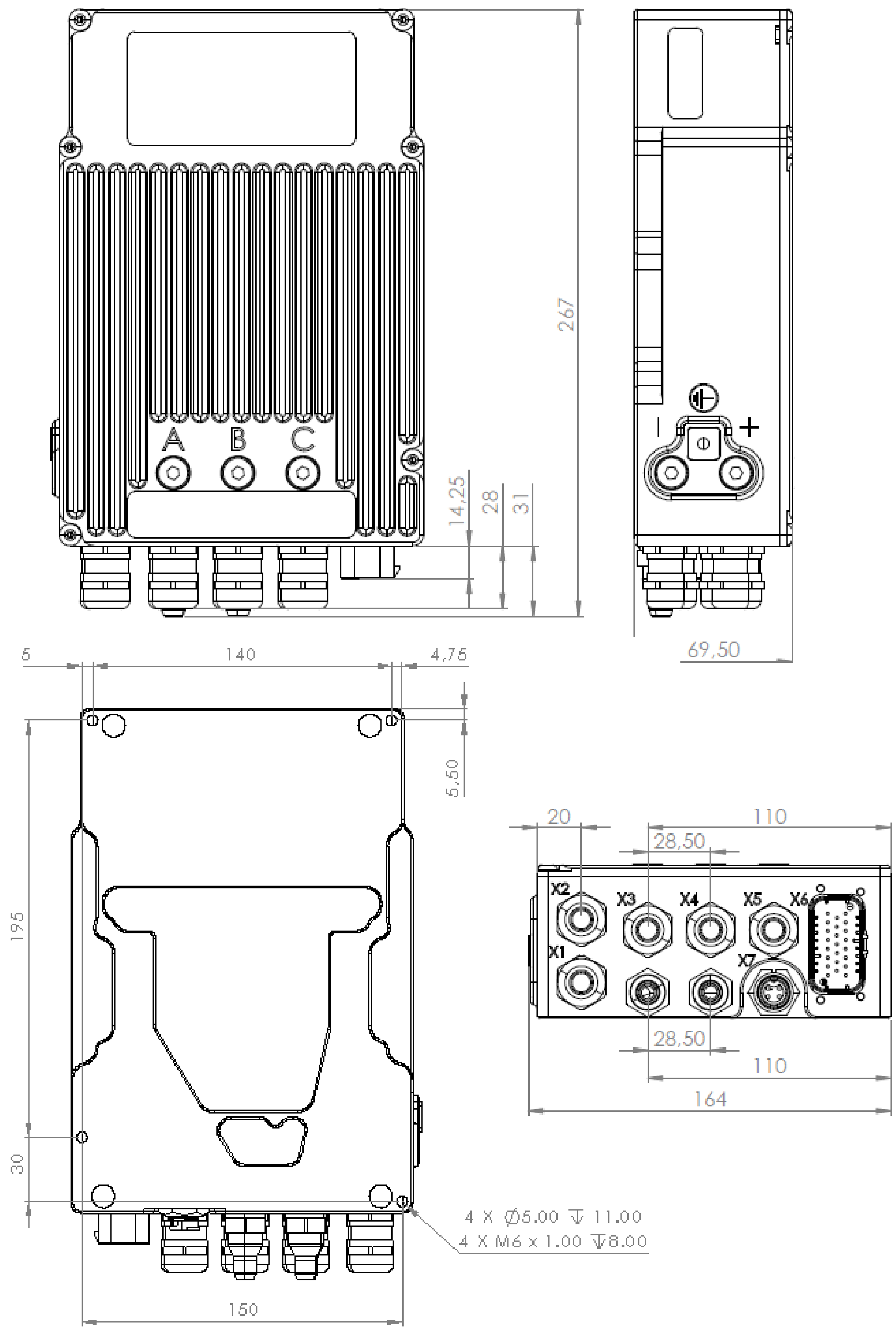




Reference	Description
1	DC HV cable connectors
2	AC HV cable connectors
3a	Temperature sensor connections
4a	Cooling connectors
5	Position, Control and LV connector
6	Earthing - Grounding connector
7	Pressure compensation plug

5.3.Dimensions

Description	Value	Unit
Mass	3.5	kg
Volume	2.8	l
Height	65	mm
Width	160	mm
Length	270	mm


267
A
B
C
14,25
28
31
69,50
5
140
4,75
5,50
195
30
4 X Ø5.00 ⊽ 11.00
4 X M6 x 1.00 ⊽8.00
150
20
110
28,50
X2
X3
X4
X5
X6
X1
X7
28,50
110
164

6. PROTECTION FUNCTIONS

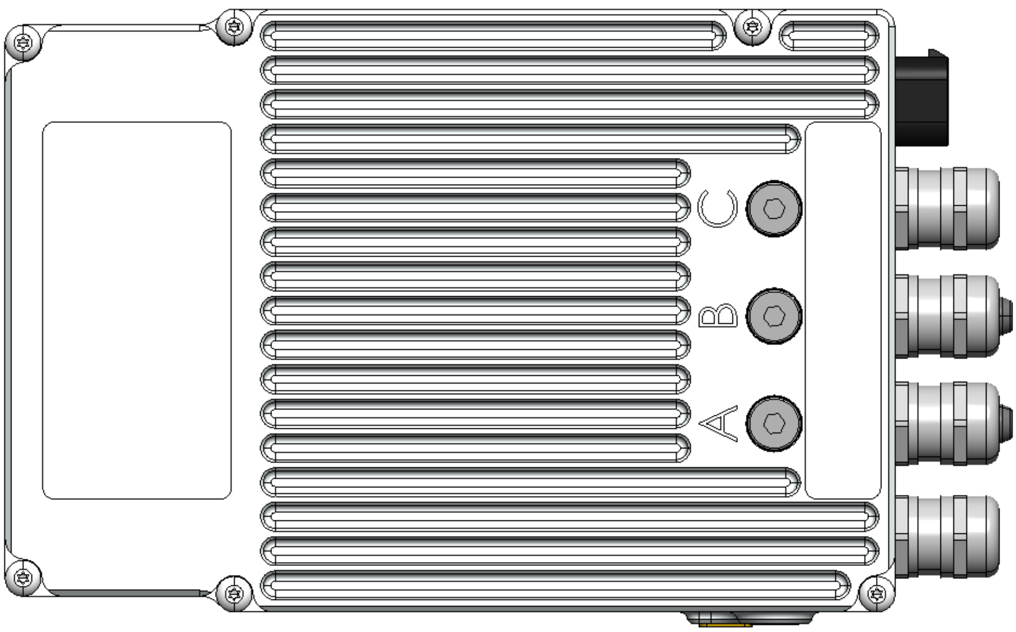
Symbol	Description	W90
OCP	HW protection detection for maximum current in the phases	640 [A]
OVP1	HW protection detection for first threshold in DC voltage	440 [V]
OVP2	HW protection detection for second threshold in DC voltage	490 [V]
OTP	HW protection detection for maximum temperature in the phases	85 [ºC]

- If OCP is violated the system goes to freewheeling mode, opening the IGBTs.
- If OVP 1st threshold is violated the system goes to freewheeling mode, opening the IGBTs.
- If OVP 2nd threshold is violated the system goes to ASC mode, closing the low side and opening the high side IGBTs not allowing any voltage.

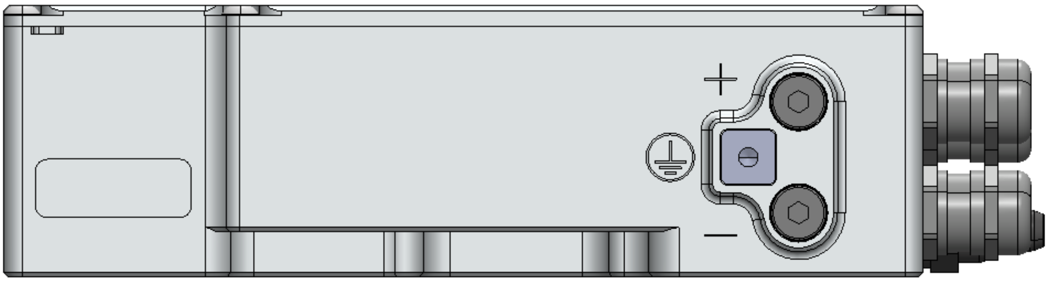
7. INTERFACES

The inverter electronics is enclosed in an Aluminum housing, which is attached by seven M5x12 screws (DIN 7380) plus one antitamper screw, which should not be released. If the customer opens the inverter unscrewing these eight screws without prior written approval from EPowerlabs, the legal warranty will be void.

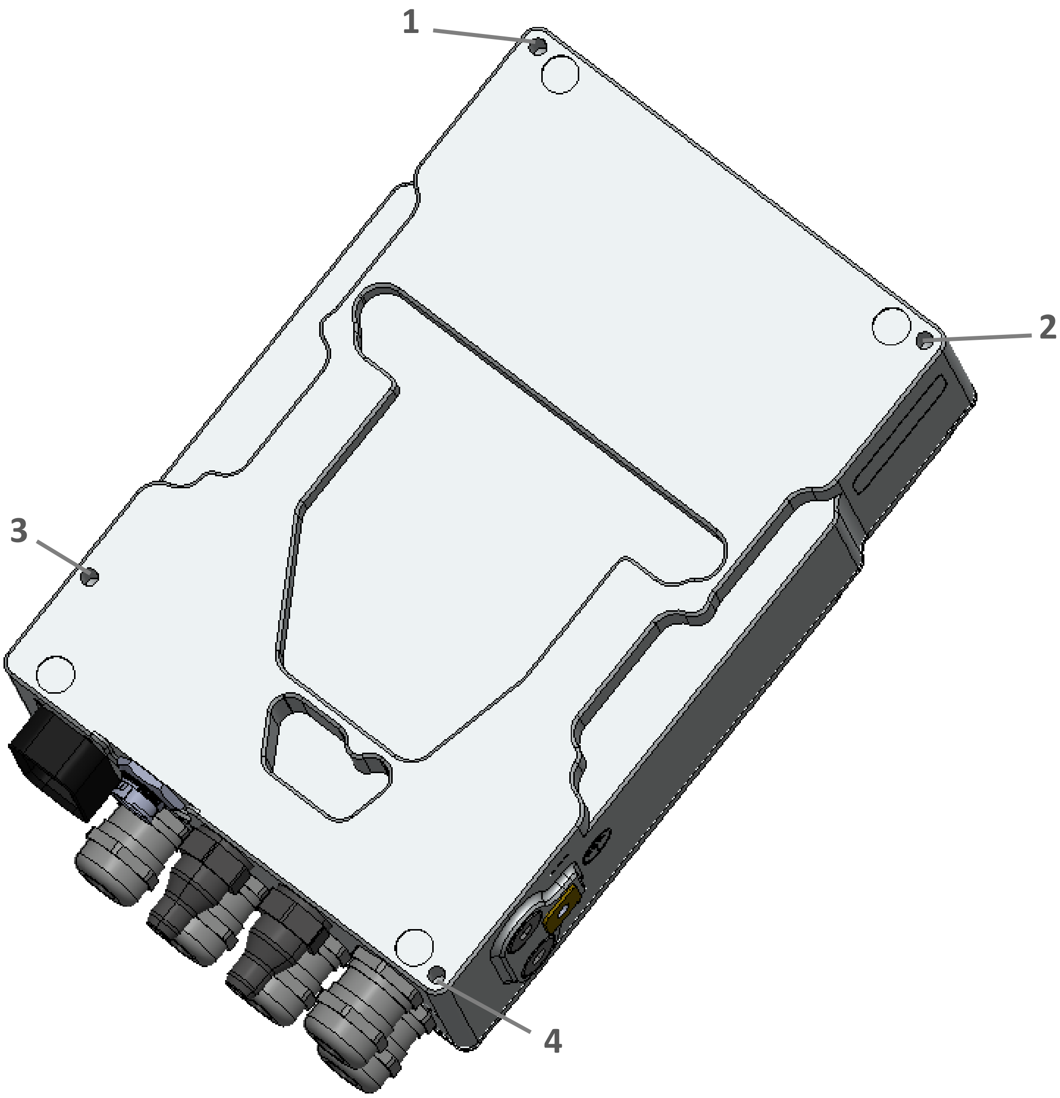
The housing has 5 extra M10 screws (DIN 908 M10x1). When these are released, M10 screws (DIN 913 M10x10) would be seen, which are the high voltage wires attachments. The manipulation of



these metric screws lets the user to connect and disconnect the power wires without the need of releasing the housing cover.



The mechanical fastening is to be aligned in such a way that the device is installed in a secured position and with as few vibrations as possible.



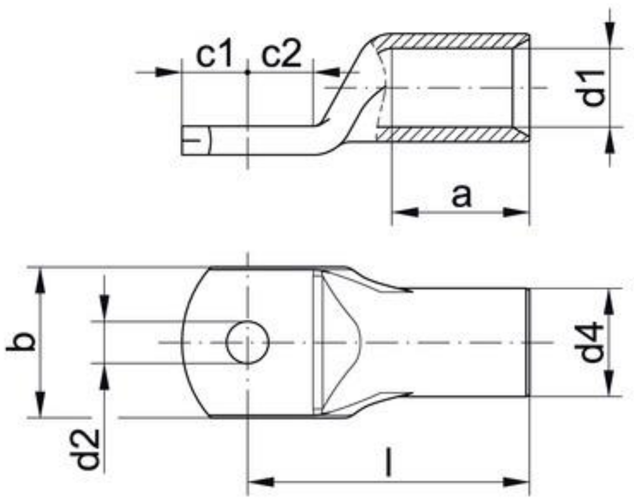
The inverter comes equipped with six holes distributed on the rear side of the housing. This makes it easy to the user to install six M6x8mm screws and fix the inverter to the desired surface.

7.1.DC & AC High Voltage connections

The default cable glands mounted on the inverter for HV connections is LAPP 53112020. This cable gland allows cables between 7 to 13 mm diameter. EPowerlabs recommends 35 mm^2 wires with this inverter. However, it is up to the final user or customer to assess what cable size should be used depending on the current levels of the target application.

The HV wires will have to be connected to a cable lug with an M8 hole to ensure a proper connection with the inverter. The suggested cable lug for 35 mm^2 wires is Klauke 398-1968. For the correct operation of the clamping system, make sure that the **c1** measurement does not exceed 7.5mm, **b** measurement does not exceed 16mm and the height of the lug is at least 2,5mm up to 3,5mm.

Alternatively, EPowerlabs can evaluate special customer requirements such as busbars or Amphenol connectors.



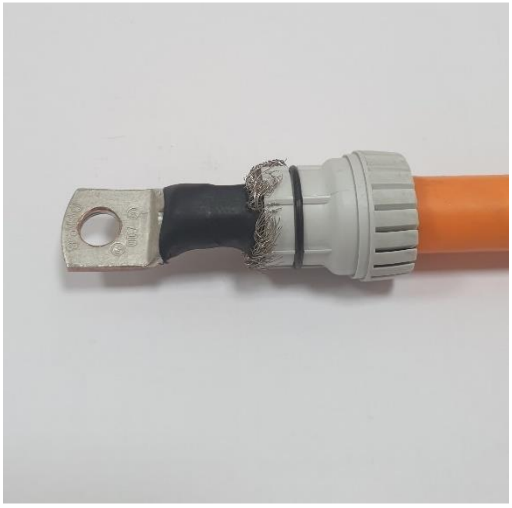
Shielded cables for the 3 AC phases are strongly recommended and full operating characteristics of the inverter are ensured only with shielded cables. Shielded cables are also recommended for the DC side, although is not as essential as with the AC output.

It is the customer´s responsibility to ensure adequate cables are used, ensure a proper connection between cable and cable lug and follow the mounting process. The following points describe the suggested HV cable assembly process and the HV cable mounting and dismounting process. These procedures try to address most common faults, but in case of doubt EPowerlabs encourages its customers to seek guidance.

7.1.1. Shielded HV cable assembly process

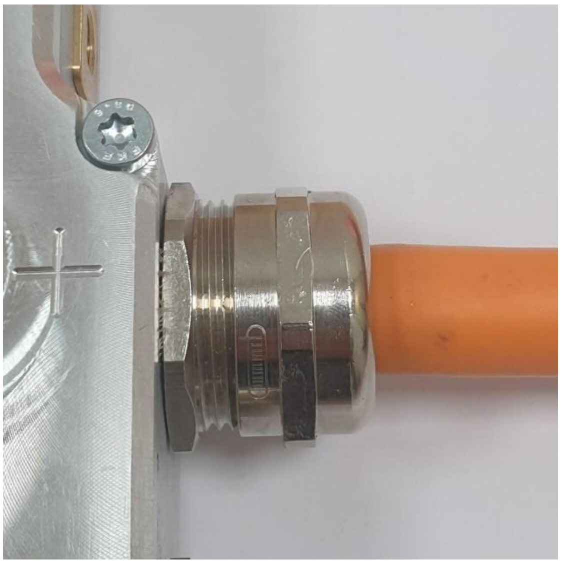
1. Cut of a section of the protective cover of the shielded cable to expose the conductive mesh or cable shield.
2. Cut of a section of the external protective cover, shield, and internal protective cover until exposing the phase copper.
3. Connect lug with M6 hole to cable copper and cover the connection with heat shrinking tape.

4. Place braided shield over the contact bushing and cut off the protruding shielding material. Secure that de shield will not exceed the O-ring.



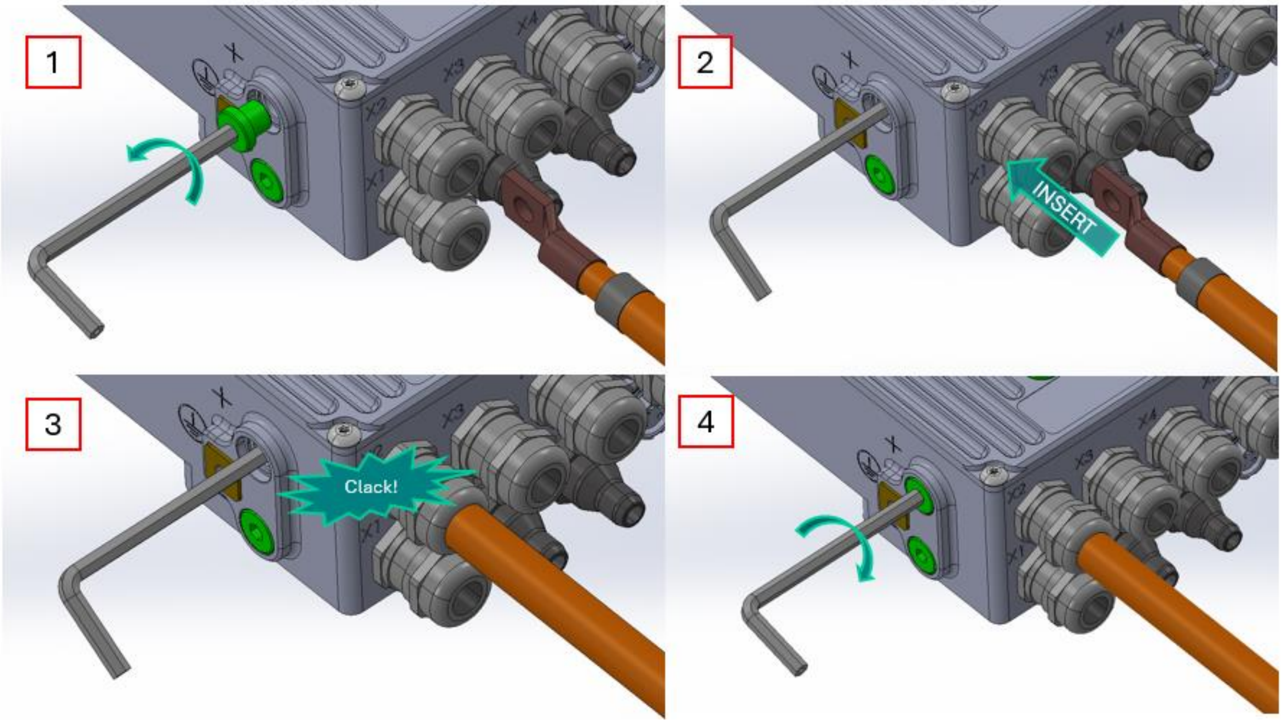
5. Push the compression nut with the contact bushing.

6. Insert in the lower part of the cable gland and tight.



7.1.2. Shielded HV cable mounting process:

1. Unscrew the M10 cover plug with a Nr.5 Allen.
2. Loosen up the M10 socket set screw with a Nr.5 Allen.
3. Loosen up the cable gland and remove cable gland plastic body.
4. Ensuring that the cable shield does not touch the phase copper nor lug, imprison the cable shield between the outer seal nut and the cable gland plastic body, without jamming the thread.
5. Insert the cable until hearing mechanical feedback.
6. Pull on the cable by hand to check that is properly connected and does not come out.
7. Tighten the M10 socket set screw. (*Tightening torque = 4Nm*)
8. Tighten the cable gland.
9. Insert and tighten the M10 cover plug.



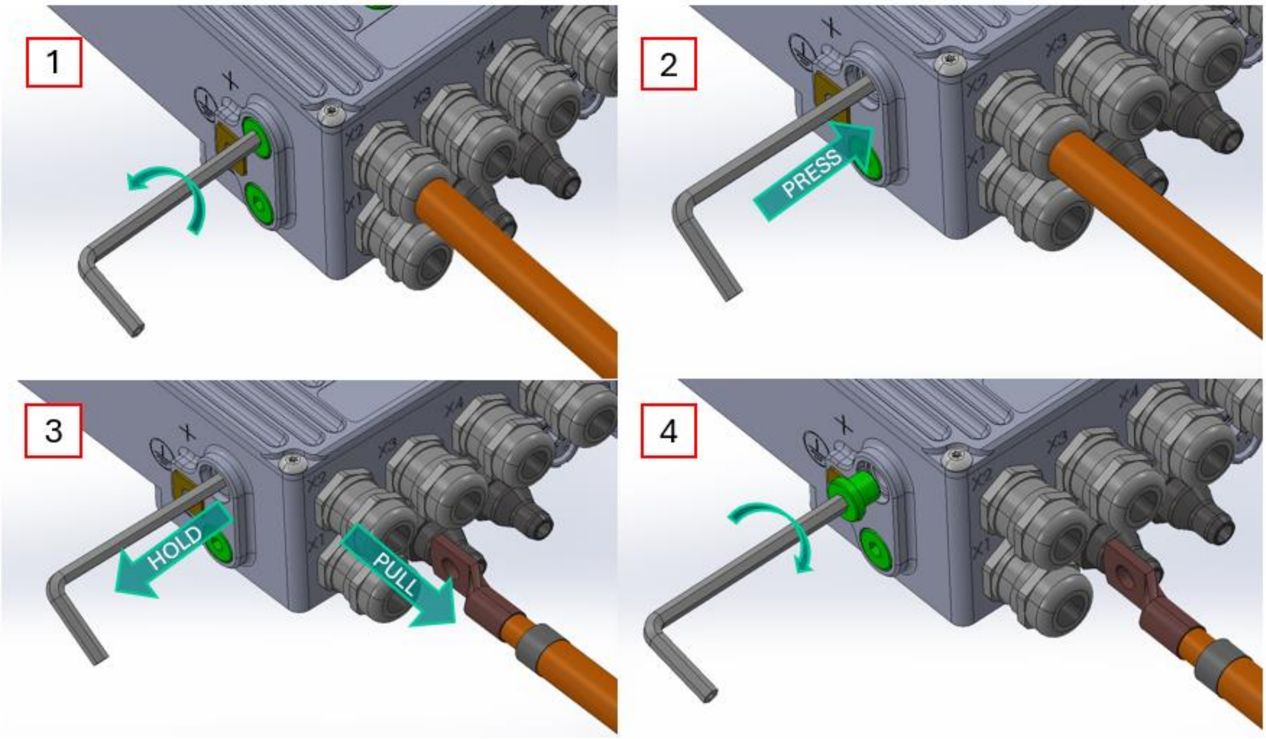
Once AC and DC cables are mounted in the inverter, test every cable for isolation with an isometer at 550V.

a) Isolation of every AC phase to ground/shield should be >10MOhms.
b) Isolation of one AC phase to every other AC phase (without mounting the motor) should be >2MOhm.
c) Isolation of every DC cable to ground/shield should be >10MOhms.
d) Isolation of one DC cable to the other DC cable should be >2MOhm once the DC link capacitor is charged.

Thereafter, mount the motor side of the AC cables with the other side of the cables shield attached to the motor body chassis and check for isolation again. Now check that 3 AC phases are connected through the motor windings and check isolation of any phase to ground/shield/motor body and should be >2MOhms.

7.1.3. Shielded HV cable dismounting process

1. Unscrew the M10 cover plug with a Nr.5 Allen.
2. Loosen up the M10 socket set screw with a Nr.5 Allen.
3. Loosen up the cable gland.
4. Press and hold the socket set screw with the Allen key to release the HV cable and pull the cable out.
5. Insert and tighten the M10 cover plug.
6. Tighten the cable gland.



7.2.Cooling interface

The default circuit is frontal cooling, where the cooling inlets are located in parallel to the HV connections.

The connector featuring on the inverter is a VDA screwed-in NW06 – M14x1.5 for Normaquick PS3 (NOR-40061174X). The counterpart is the Normaquick PS3 NW06. These fittings are available in several angles.

7.3. Low Voltage Connections

7.3.1. CONTROL CONNECTOR (X6)

The inverter features a AMPSEAL 23 pos 1-776228-1 (X6), which mates with AMPSEAL 770680-1 (770520-5 terminal). To better ensure wiring harness integrity two backshell pieces can be placed in order to protect the connector terminal end (2389806-1), in case one or several pins are left unused to ensure full water tightness sealing plug (770678-1) can be used in the unpopulated terminal pins. For further assembly instructions refer to the following link [AMPSEAL Connector Overview & Assembly Instructions].

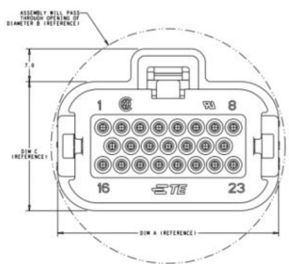
This connector drives the power supply of the control side of the inverter and the communications among other control signals.

The control part of the device is needs a 12VDC power supply to work, the operation range being between 9V and 28V, with a 2 A maximum current.

It is strongly recommended to use shielded and twisted wires for CAN lines, 5-6 and 9-10 pins as identified on the following table.

Wires from connector to the position sensor must be twisted in pairs and it is strongly recommended to use shielded cables for all connections. The wires should be AWG22.

Pins 16 to 19 are common for resolver and xMR, 14 and 15 are specific for resolver and 20 and 21 are specific for xMR.



Description	Value	Unit
Resolver excitation signal at 10kHz	7	Vpp
Resolver excitation resistance	75	Ohms
Resolver Sinus-Cosinus output voltage	1 - 4	Vpp
xMR/Encoder supply	5	V
xMR/Encoder max current	50	mA

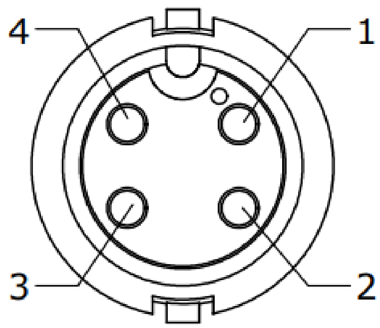
Pin	Name	Description	Wire thickness
1	KL30	Positive battery	AWG20
2	KL31	Negative battery	AWG20
3	KL15	Ignition	AWG22
4	KL31	Negative battery	AWG20
5	CAN1_H	CAN1 High 120Ω terminated	AWG22
6	CAN1_L	CAN1 Low 120Ω terminated	AWG22
7	HVIL_IN	Interlock IN	AWG22
8	KL30_AUX	Positive battery auxiliary	AWG20
9	CAN2_H	CAN2 High 120Ω terminated	AWG22
10	CAN2_L	CAN2 Low 120Ω terminated	AWG22
11	BOOT	Bootloader lock	AWG22
12	HVIL_OUT	Interlock OUT	AWG22
13	EXC_P	Excitation positive	AWG22
14	EXC_N	Excitation negative	AWG20
15	POS_C	Encoder C	AWG22
16	X_N	Cosine negative	AWG22
17	X_P	Cosine positive	AWG22
18	Y_P	Sine positive	AWG22
19	Y_N	Sine negative	AWG22
20	5V	5V positive supply	AWG22
21	GND	5V negative supply	AWG22
22	POS_A	Encoder A	AWG22
23	POS_B	Encoder B	AWG22

7.3.2. TEMPERATURE SENSOR CONECTOR (X7)

This temperature sensor connector featuring on the inverters is a Switchcraft MINI-CON-X 7261-4SG-300 (X7), which would require its mate Switchcraft MINI-CON-X 6282-4PG-315-CC1.

Wires from connector to the sensors must be twisted in pairs. The wires should be AWG22.

The inverter is designed to work with NTC 10k temperature sensors, the acceptance range being between 1k and 100k. Alternatively, the inverter can work with PT1000 sensor or equivalent KTY. If a client requires another or a combination of other temperature sensors, EPOWERLABS will evaluate the change request.



**Seen from outside of Inverter*

Pin	Name	Description
1	NTC2_N	NTC2 negative (ISOGND)
2	NTC2_P	NTC2 positive
3	NTC1_N	NTC1 negative (ISOGND)
4	NTC1_P	NTC1 positive

7.4.Communication protocol

The inverter features 2 CAN communication channels:

- Channel 1: CAN 2.0B or CAN-FD with 500 kbit/s baud rate for vehicle ECU communication. Baud rate of Channel 1 is adjustable upon customer request.
- Channel 2: CAN 2.0B with 1 Mbit/s baud rate to connect DeveLinkSTUDIO calibration and debug tool.

The .dbc is described in detail in ANNEX 1: Default DBC.

7.5.Sourcing of auxiliary connectors

For the customer to have access to DeveLinkSTUDIO user interface, it requires a PCAN-USB-FD adaptor from PEAK systems with a PCAN-Term. Epowerlabs can supply these alongside the inverter at a cost.

Likewise, Epowerlabs can at a cost if requested by the customer supply the following LV connectors with cables pre-assembled:

- Control connector: AMPSEAL 770680-1 with 2m cables ended by DB9's and bananas.
- Temperature connector: 6282-4PG-315 SWITCHCRAFT with 2m open end cables.
- Cooling connector: Normaquick PS3 NW6 0º (2 per inverter)

8. Product Warranty

The product warranty is not applicable to prototype or pre-series samples.

The product warranty is established with each customer as part of the commercial offer. In any case, the product warranty only covers defects attributable to the design or manufacturing of the product. EPowerlabs will repair o substitute the defective component given the customer returns it. Alternatively, Epowerlabs may offer a replacement inverter, offer a discount or a full reimbursement.

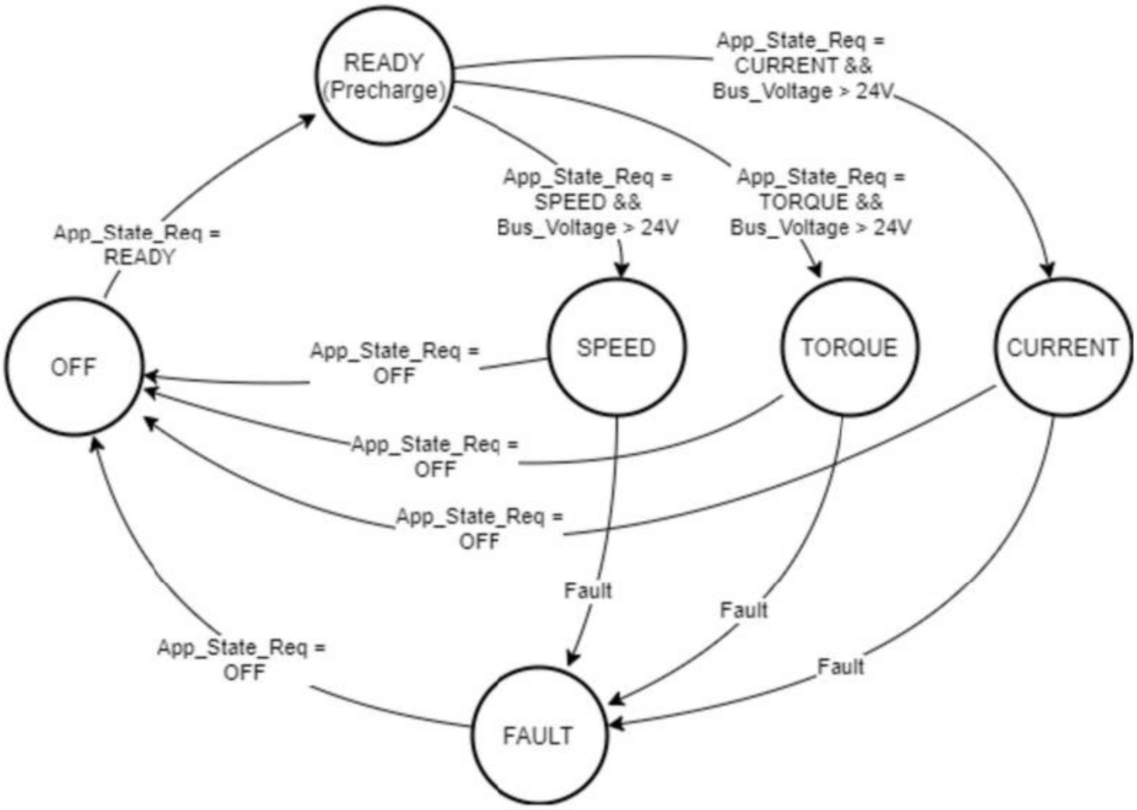
The product warranty does not cover any indirect damages or costs due to production stoppages or inactivity.

The product warranty will be void if:

- The customer has not adhered to the guidelines stated in the user manual or other written guidelines provided by the seller.
- If the inverter has not been used for its intended purpose without prior written acknowledgement from EPowerlabs.
- The inverter has been required to operate outside the operational range stated in the user manual.
- The inverter case been opened or corrupted in any form.
- A component of the inverter has been substituted without prior written acknowledgement from EPowerlabs.
- Damages for reasons beyond EPowerlabs´s control.
- A customer may request Epowerlabs the .hex and .elf files to have unrestricted access to the calibration software. If access to the .hex and .elf files is provided, any warranty claim will be void and Epowerlabs will not take any responsibility to damages caused to the inverter and other equipment caused by the inverter thereafter.
- The user manual will be facilitated to the client alonside the Purchase Order Confirmation. If between the Purchase Order and the product delivery, there would be a new version of the User Manual relevant to the product being supplied, Epowerlabs will endeavour to facilitate access to the most relevant User Manual version. In case of doubt, it is the client's responsibility to contact info@epowerlabs.com to enquire about new versions of the user manual.

9. SYSTEM INTEGRATION AND START UP

9.1. State machine



9.2. Faults

9.2.1. L1: Power stage

This is the feedback signal read from the hardware protection, wich maintains freedom of interface.

Bit	Value	Name	Description	Safe state
1	1	Alive	Hardware interface is funcional	None
2	2	Enable	Power stage is enabled	None
3	4	UVLO	Undervoltage (primary or isolated side)	None
4	8	Desat	Desaturation/Overcurrent	Freewheeling
5	16	DT_Violation	Deadtime violation	Freewheeling
6 7	32 64	HVIL_Open OCP	High voltage interlock fault Phase overcurrent	Freewheeling Freewheeling
8 9	128 256	OVP_TH1 OVP_TH2	Phase overvoltage – threshold 1: Freewheeling Phase overvoltage – threshold 2: ASC	Freewheeling Active Short Circuit

9.2.2. L2: Electric Machine Control

This layer is dedicated to EMCtrl and its children modules: EMCtrl_DC, EMCtrl_Blk and EMCtrl_FOC.

Bit	Value	Name	Description	Safe state
1	1	Init OK	Software module initialized	None
2	2	PosFb	Position feedback fault	Freewheeling
3	4	ASC	Software active short circuit	Active Short Circuit
4	8	Curr_Imbalance	Phase current imbalance	Freewheeling
5	16	PwrStg_Fault	Hardware fault (See L1 faults)	Freewheeling
6	32	Current_derating	Q setpoint removed to maintain voltage limit	Freewheeling
7	64	Loop_delocked	Unable to reach a stable current control	Freewheeling
8	128	PhCurrAcqFault	Phase current acquisition fault	Freewheeling

9.2.3. L3: Diagnostic Event Manager

The Diagnostic Event Manager or DEM, is the top level diagnostic backbone for the whole embedded application.

Code	Description	Safe state
1	Lost msg	Freewheeling
2	Undervoltage	Freewheeling
3	PwrStg Overtemperature	Freewheeling
4	PwrStg temperature degradation	None
5	EMCtr fault	Freewheeling
6	Task overrun	Freewheeling
7	CAN1_BusOFF	Freewheeling
8	Emachine_overtemp	Freewheeling
9	Phase Current sensor out of range	Freewheeling
10	PwrStg temperature sensor out of range	Freewheeling
11	DCBus voltage sensor out of range	Freewheeling
12	DP board overtemp	Freewheeling
13	DRV board overtemp	Freewheeling
14	Aux supply out of range	Freewheeling
15	Emachine_overspeed	Freewheeling

9.3.Start UP guide.

- KL30 and KL15 to 12V to start up the inverter.
- Default inverter command source App_TST (CAN1)
- **CAN1 ECU communication** (Annex 1 DBC) state machine
- App_State_Req: 1.OFF 2. READY 3. CURRENT 4. TORQUE 5. FAULT
- For the inverter to go from 2. READY to 3. CURRENT the correct DC precharge has to be performed and the correct I_D/I_Q setpoints have to be set (Recommended every 10ms). Going to 1. OFF should restart any 5. FAULT or CAN1 timeout errors.
- **For DevelinkSTUDIO operation** a separated chapter 8 is used for describing the operation.describing the operation.

10. DeveLinkSTUDIO Quick Guide and Calibration.

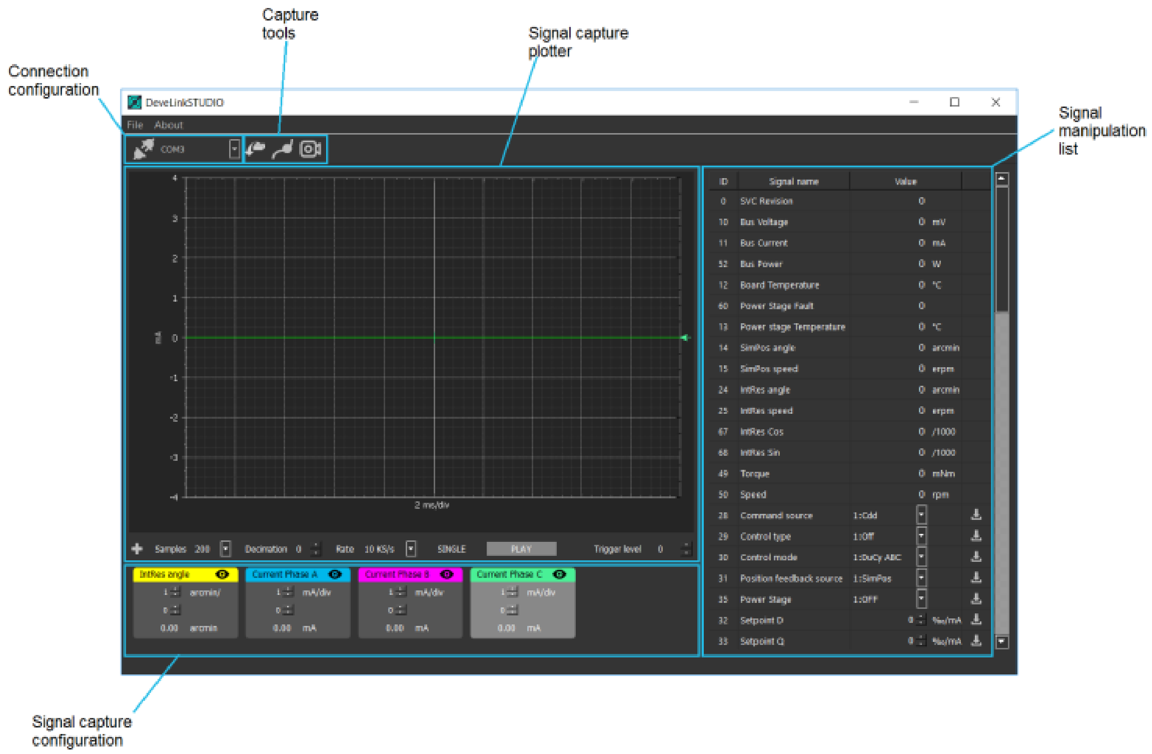
The EMC 150 offers an own software developed by EPowerlabs: DeveLinkSTUDIO.

10.1. Quick Guide.

When DeveLinkSTUDIO is started, the main window appears, which contains different areas (clockwise in the picture):

- Connection configuration (toolbar): configures the serial port it should communicate to (Recommended to use Peak Systems IPEH-002022)
- Capture tools (toolbar): additional tools for visualization/post-processing
- Signal capture plotter: Plots the configured signals
- Signal manipulation list: Gets/sets the value of the listed signals
- Signal capture configuration: Configures the signal to be captured on the specific channel.

Adapts the scale/offset of the visualization and, in Continuous capture mode (explained later), calculates the average value.

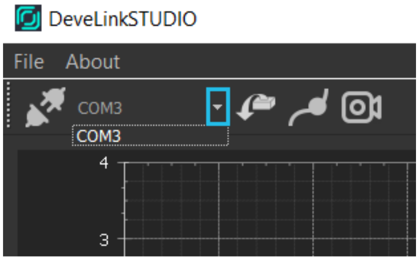


For the signal acquisition, three parameters can be configured:

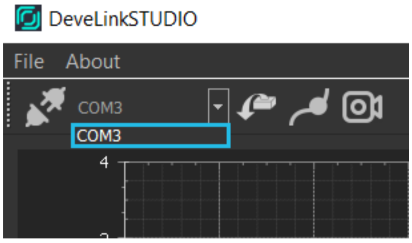
- Samples (Only in buffered mode): Number of points to store in the buffer
    - 100, 200, 500 or 1000 sample points
- Decimation (Only in buffered mode)
    - Undersampling with averaging of the signal
- Acquisition modes: In both modes, the capture is synchronized, meaning that, there is no shift.
- in the signals during capture, all 5 signals samples have the same timestamp.
    - Buffered (10, 20 or 40Khz sampling frequency)
    - Continuous (10ms period)
- Capture state: Capture can be paused (continuous and buffered mode) and single-triggered.
- (buffered)
    - Play/stop button
    - Single capture mode (only in buffered mode)
- Trigger level: The firsts signal (yellow) defines the trigger value. The level of the trigger can be configured via this setting. It is a rising type trigger.

SET UP A CONNECTION

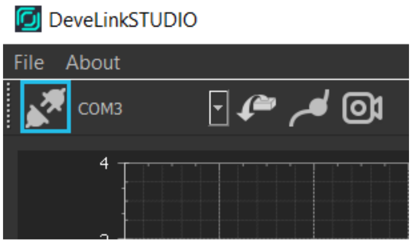
Once the computer running DeveLinkSTUDIO is physically connected to the embedded system using two of the known interfaces (UART/USB or CAN), the connection can be established. First, select the serial port the device is connected to. This can be found by checking the list of serial ports before and after the device is plugged (list is updated every time the drop-down arrow is clicked.



Select the right COMx port

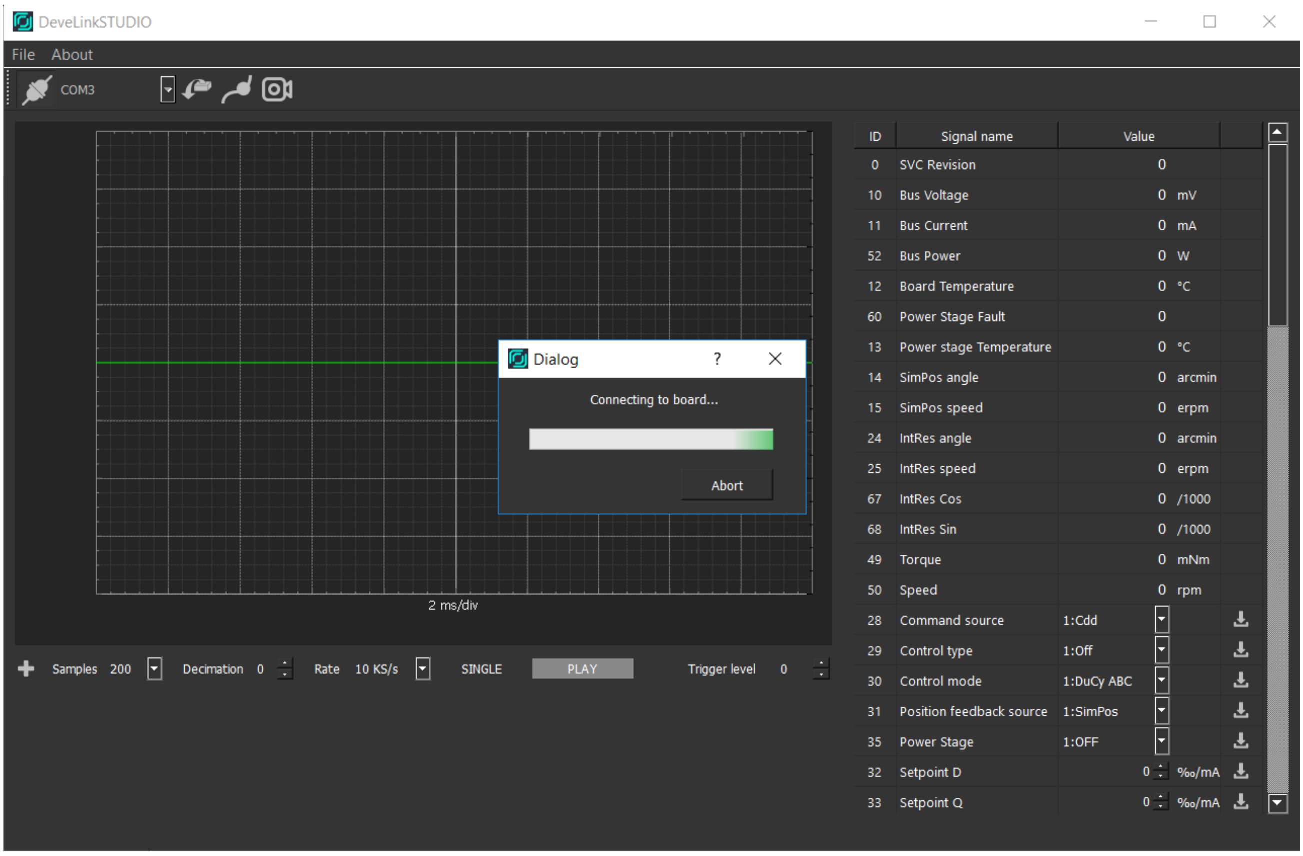


Click on the “Connect to device” button

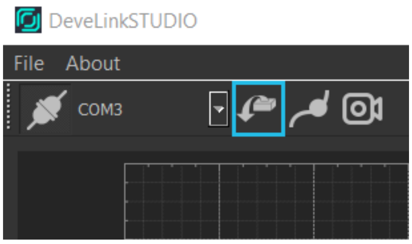


Once the serial port is open, it will attempt to connect to the embedded platform. A window will briefly appear. If everything succeeds, it will be ready to communicate. Should the window stay showing “Connecting to board…”, there is an issue with the connection. Some troubleshooting steps might be:

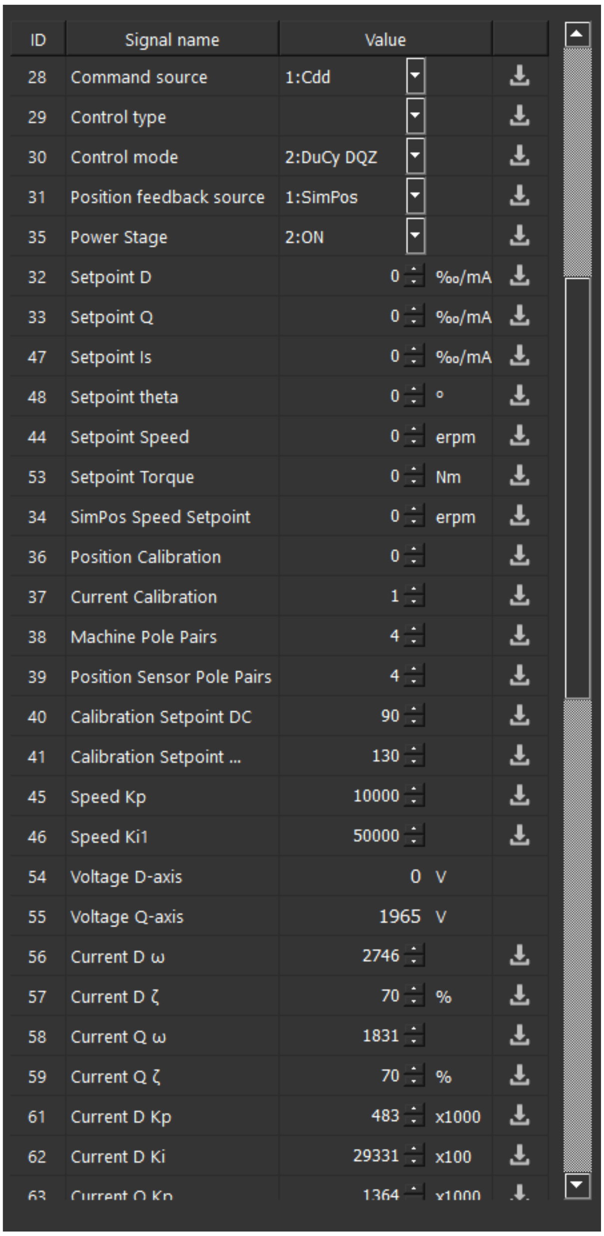
- Re-connect USB
- Reset embedded platform
- Check the environment (noise levels, cables…)



Should the connection succeed, the signals from the board can be read and the status con the controller assessed.



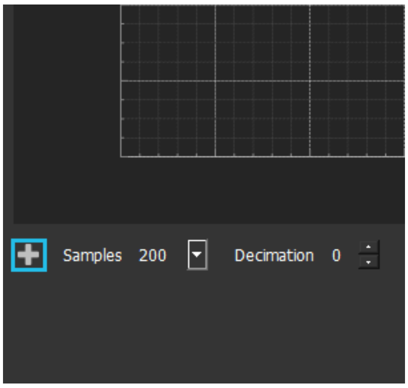
The list of signals should be updated with the values from the platform.



ID	Signal name	Value	
28	Command source	1:Cdd	
29	Control type		
30	Control mode	2:DuCy DQZ	
31	Position feedback source	1:SimPos	
35	Power Stage	2:ON	
32	Setpoint D	0	‰/mA
33	Setpoint Q	0	‰/mA
47	Setpoint Is	0	‰/mA
48	Setpoint theta	0	°
44	Setpoint Speed	0	erpm
53	Setpoint Torque	0	Nm
34	SimPos Speed Setpoint	0	erpm
36	Position Calibration	0	
37	Current Calibration	1	
38	Machine Pole Pairs	4	
39	Position Sensor Pole Pairs	4	
40	Calibration Setpoint DC	90	
41	Calibration Setpoint ...	130	
45	Speed Kp	10000	
46	Speed Ki1	50000	
54	Voltage D-axis	0	V
55	Voltage Q-axis	1965	V
56	Current D ω	2746	
57	Current D ζ	70	%
58	Current Q ω	1831	
59	Current Q ζ	70	%
61	Current D Kp	483	x1000
62	Current D Ki	29331	x100
63	Current Q Kp	1364	x1000

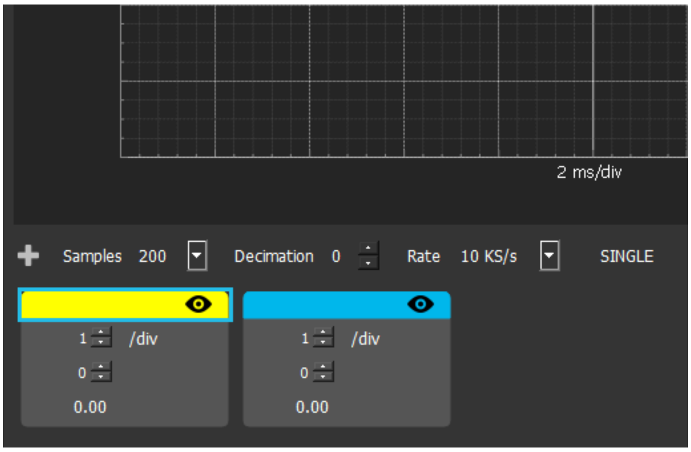
CONFIGURE CAPTURE

One can add up to 5 signals to be captured by the embedded platform and be transferred to DeveLinkSTUDIO. Once step 1.1 is performed, captures can be added by clicking on the "+" button.

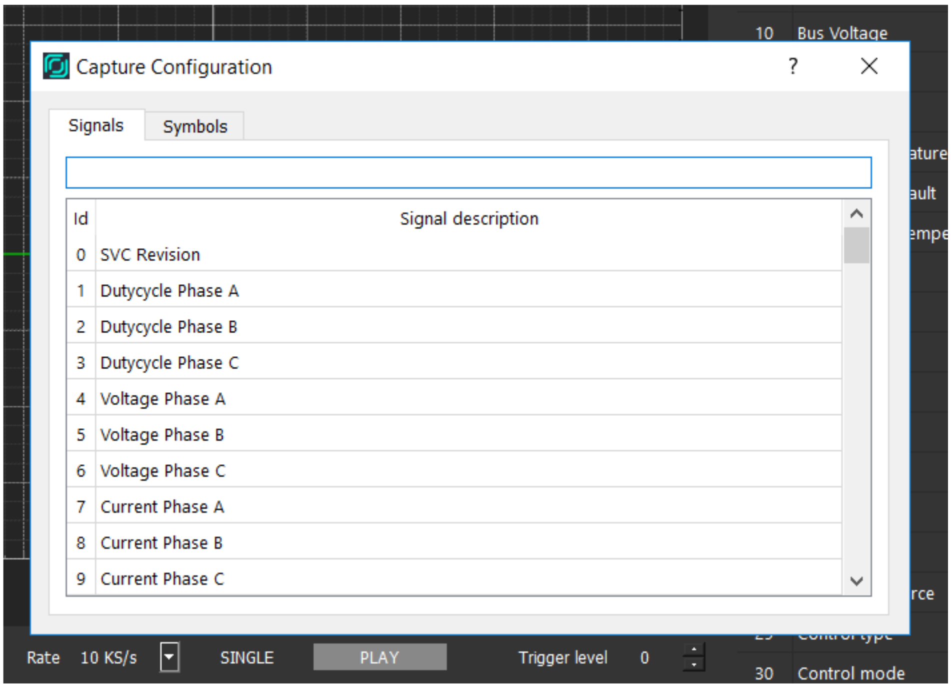


Capture configuration boxes will start to appear on the lower side of the UI.

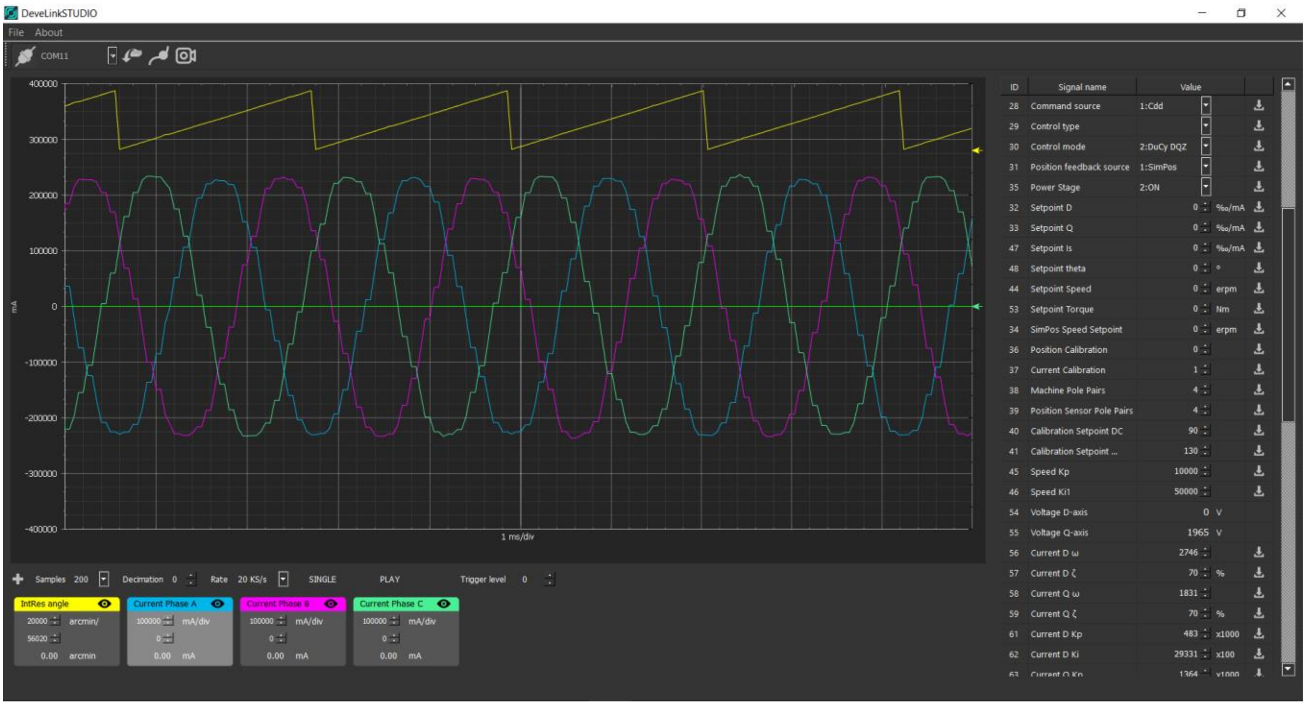
To configure the capture, double click on the colored bar of the box.

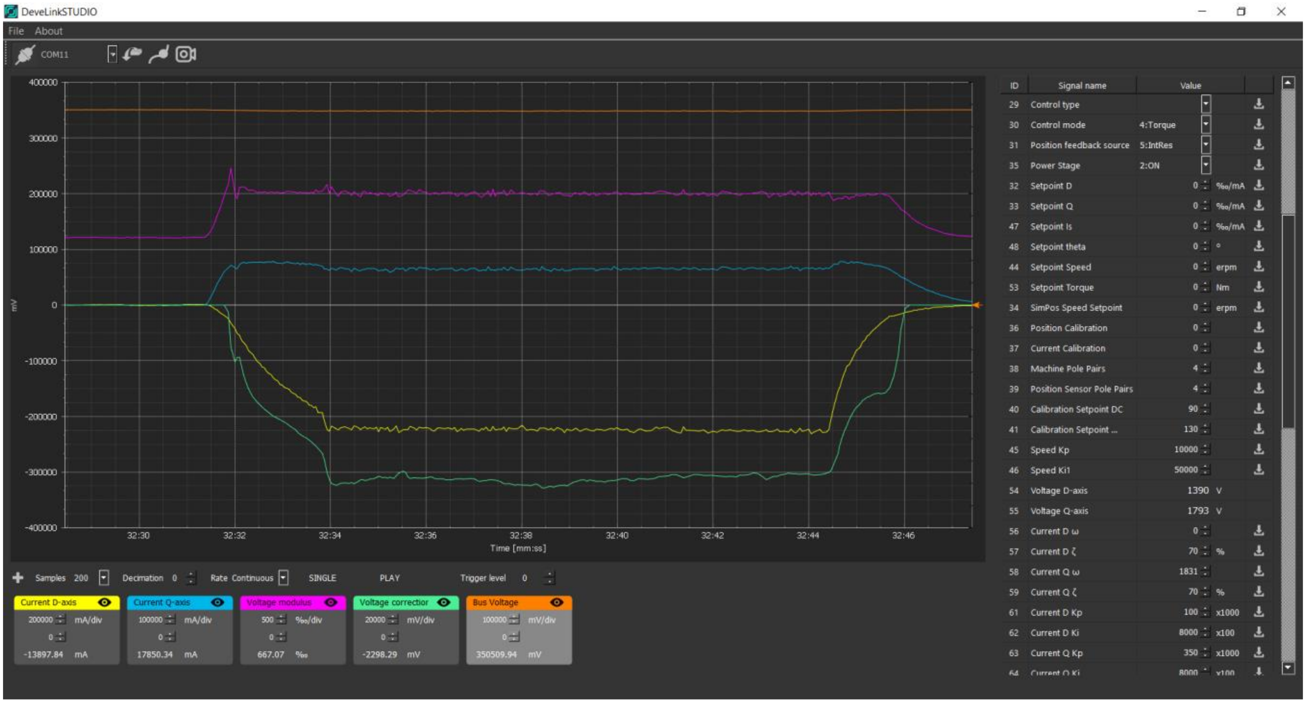


A list with all the available signals will appear. The search bar can be used to look for a particular keyword among the available signals.



Should it not be connected to any platform, a warning message will appear else, the signal will be ready to trigger (if in buffered mode) or to continuously sample (if in continuous mode).

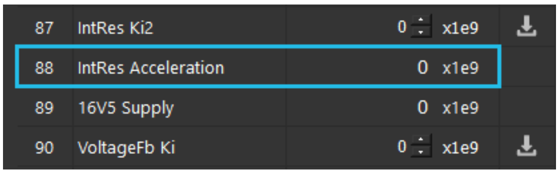




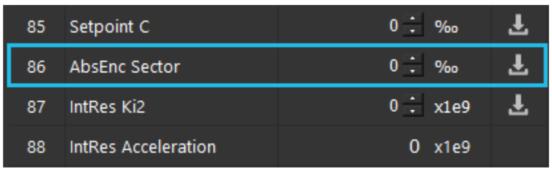
SIGNALS MANIPULATION

DeveLinkSTUDIO is also able to manipulate the signals listed in the right side of the UI. There are three types of signals:

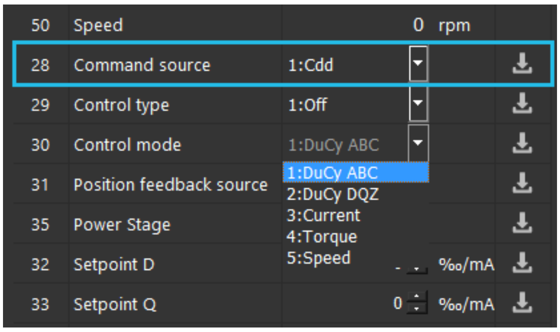
- Read-only values : signals can only be read



- Read/Write signal : any value can be set. Press enter in order to confirm the value.



- Read/Write enumerated signal : Specific values can only be selected



-

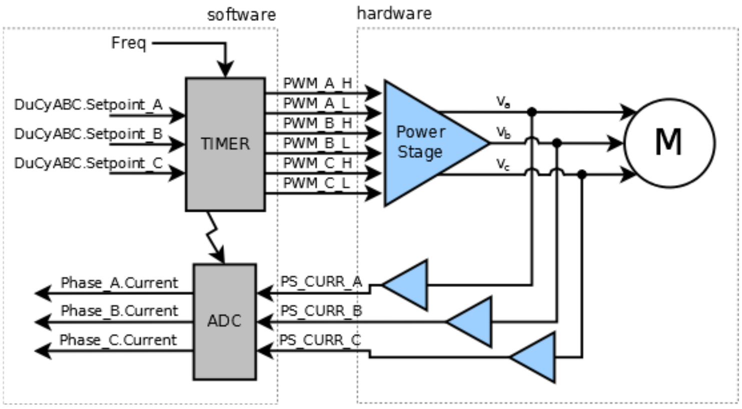
Once the value is set, it must be written on the board. This value does not write permanently. Restarts after every reset.

76	IntRes Gain Cos	0
77	IntRes Gain Sin	0
78	IntRes Delay Carrier	0
79	IntRes Delay Angle	0

Using this software, the user can perform calibration and testing process applying different control techniques, which are presented below.

10.2. OPERATION MODES

10.2.1. Voltage (PWM) operation



In this mode (*DuCyABC*), the controller acts as a three channel, center-aligned, complementary, PWM generator. This is useful for electric machine diagnostics (phase resistance/inductance, imbalance…), short-circuit testing and inverter diagnostics.

The inputs for this mode are the PWM frequency (same for all channels) and the duty cycle for each one of them. To select the mode. The following signals must be set:

- Command source: Debug
- Control type: Vectorial
- Control mode: DuCyABC
- Power Stage: ON

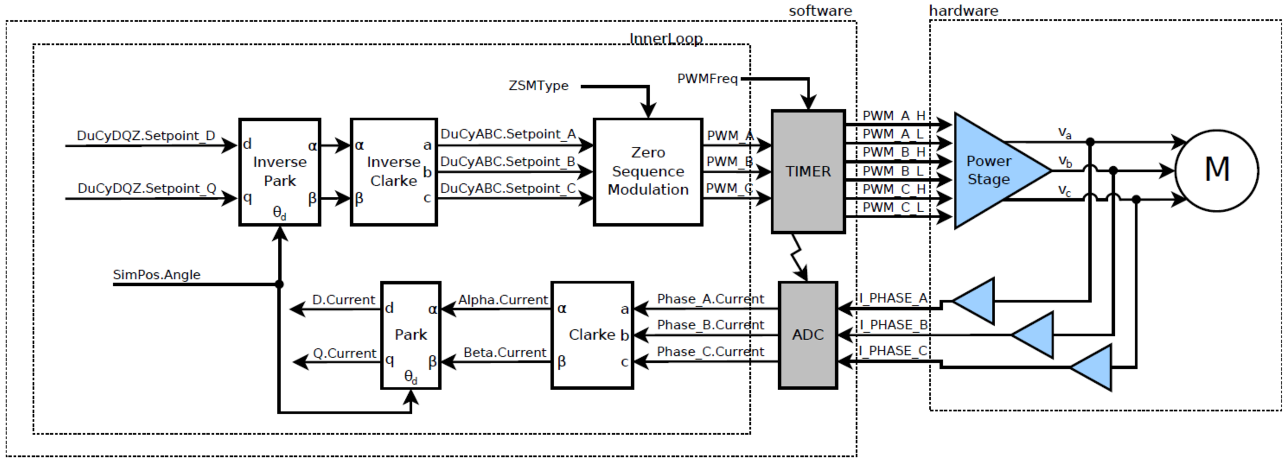
34	Command source	1:Debug
35	Control type	3:Vectorial
36	Control mode	1:DuCy ABC
37	Position feedback source	1:SimPos
38	Power Stage enable	2:ON

The position feedback source is irrelevant for operation under this mode. To control operation under this mode, Setpoint A, Setpoint B and Setpoint C are the signals to be used. Depending on the controller being used (deadtime, bootstrap...), the lower/upper setpoints might be not settable/reachable.

39	Setpoint A	0	‰
40	Setpoint B	0	‰
41	Setpoint C	0	‰

The signals have no software filter/protection. Large currents might appear, provoking very high switching losses that can destroy the controller, with the temperature protections not being able to detect the condition fast enough (energy release too quick).

10.2.2. Open-loop position, Voltage control operation



In this mode (DuCyDQZ), the controller is able to generate a rotating three-phase voltage, using a simulated position and/or speed (SimPos) as reference.

This mode is normally used to assess resolver signals quality and look for the parameters of the auto-calibration routine. Normally, current setpoints are not yet used at this stage.

34	Command source	1:Debug
35	Control type	3:Vectorial
36	Control mode	2:DuCyDQZ
37	Position feedback source	1:SimPos
38	Power Stage	2:ON

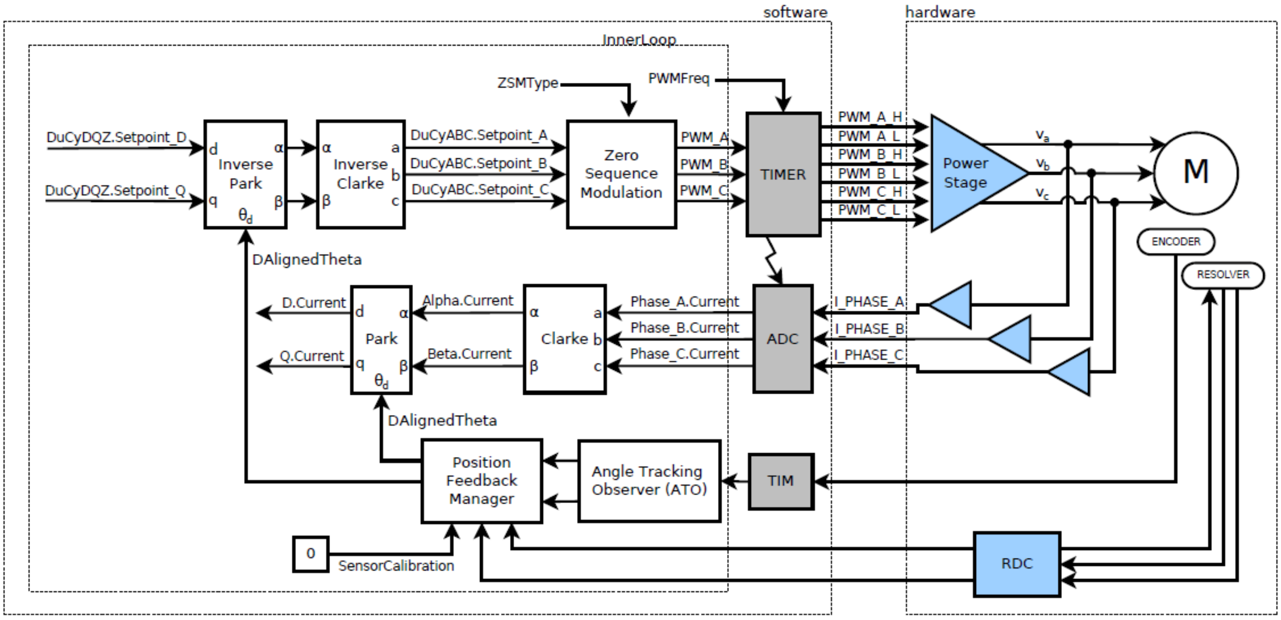
The signals used on this mode are the D/Q setpoints (cartesian) or the Is/θ (polar). Depending on the controller being used (deadtime, bootstrap...), the lower/upper setpoints might be not settable/reachable.

42	Setpoint D	0	‰/mA
43	Setpoint Q	0	‰/mA
44	Setpoint Is	0	‰/mA
45	Setpoint θ	0	°

This setpoint signals are shared with Current Control mode.

10.2.3. Closed-loop position, Voltage control operation

Position feedback is expected to be calibrated (either manually or automatically).



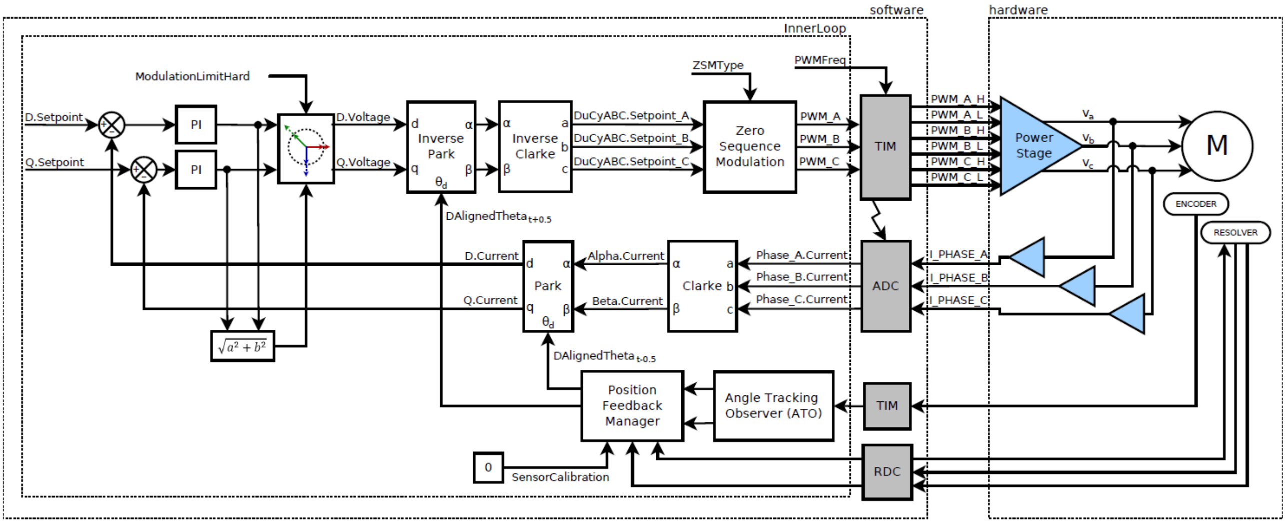
This mode (DuCyDQZ + AbsEnc/IncEnc/IntRes/xMR) is used to check position calibration by setting a voltage setpoint on Q-axis. Additionally, current readings can also be assessed. The rotating speed will be approximately proportional to the voltage set in Q-axis.

34	Command source	1:Debug
35	Control type	3:Vectorial
36	Control mode	2:DuCyDQZ
37	Position feedback source	5:IntRes
38	Power Stage	2:ON

The controller is able to set the actuation setpoint in the D/Q frame, so setpoint in D and/or Q axis (cartesian) or Is/θ (polar). Depending on the controller being used (deadtime, bootstrap...), the lower/upper setpoints might be not settable/reachable.

42	Setpoint D	0	‰/mA
43	Setpoint Q	0	‰/mA
44	Setpoint Is	0	‰/mA
45	Setpoint θ	0	°

10.2.4. Closed-loop position, Current control operation



This mode is normally used in the testbench, when a speed can be maintained. It is used to check the accuracy of the trajectory maps by setting a setpoint in D/Q or I/theta.

34	Command source	1:Debug
35	Control type	3:Vectorial
36	Control mode	3:Current
37	Position feedback source	5:IntRes
38	Power Stage	2:ON

Also, this mode can be used to assess the proper position calibration (at low speeds, were mechanical/iron losses can be neglected). Any setpoint on D should not generate any torque.

42	Setpoint D	0	‰/mA
43	Setpoint Q	0	‰/mA
44	Setpoint Is	0	‰/mA
45	Setpoint θ	0	°

10.1. SENSOR CALIBRATION

10.1.1. Resolver

To calibrate the resolver, it is necessary to be connected to the inverter. These are the steps we follow:

1. Leave the machine rotating freely.

   Command source: App_Dbg
   Control type: Vectorial
   Control mode: DuCyDQZ
   Position feedback source: SimPos
   Power Stage: ON

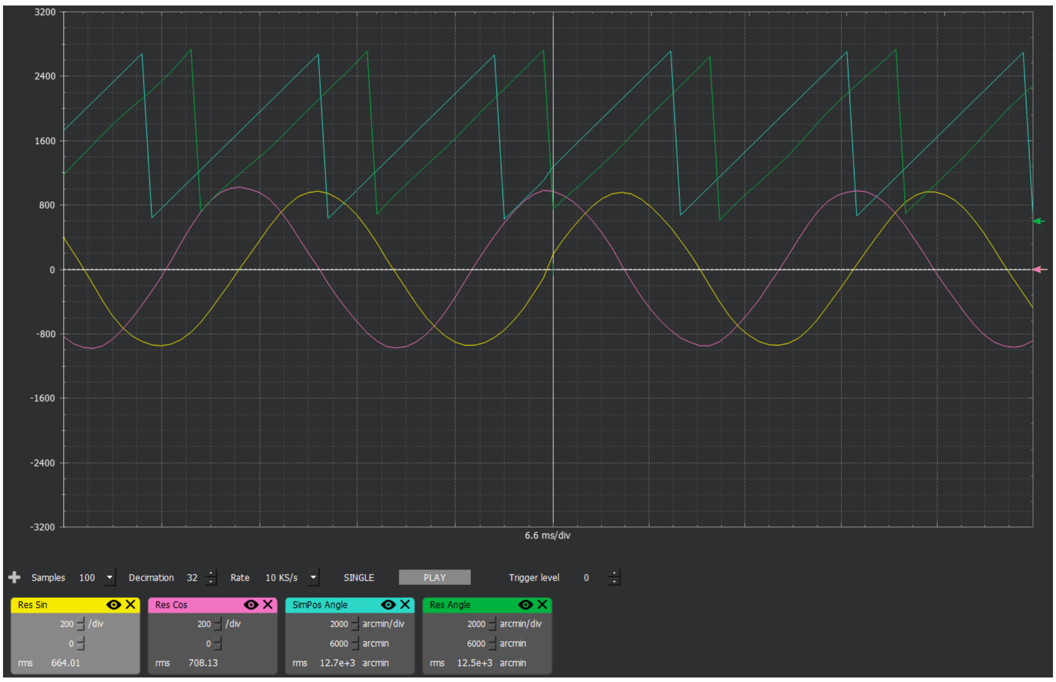
34	Command source	1:App_Dbg
35	Control type	3:Vectorial
36	Control mode	2:DuCyDQZ
37	Position feedback source	1:SimPos
38	Power Stage	2:ON

   Set the SimPos Speed to 100erpm and increase the Setpoint D in steps of 5 permil. Once you see the machine spin smoothly (relatively), you can already start tuning offset and gain.

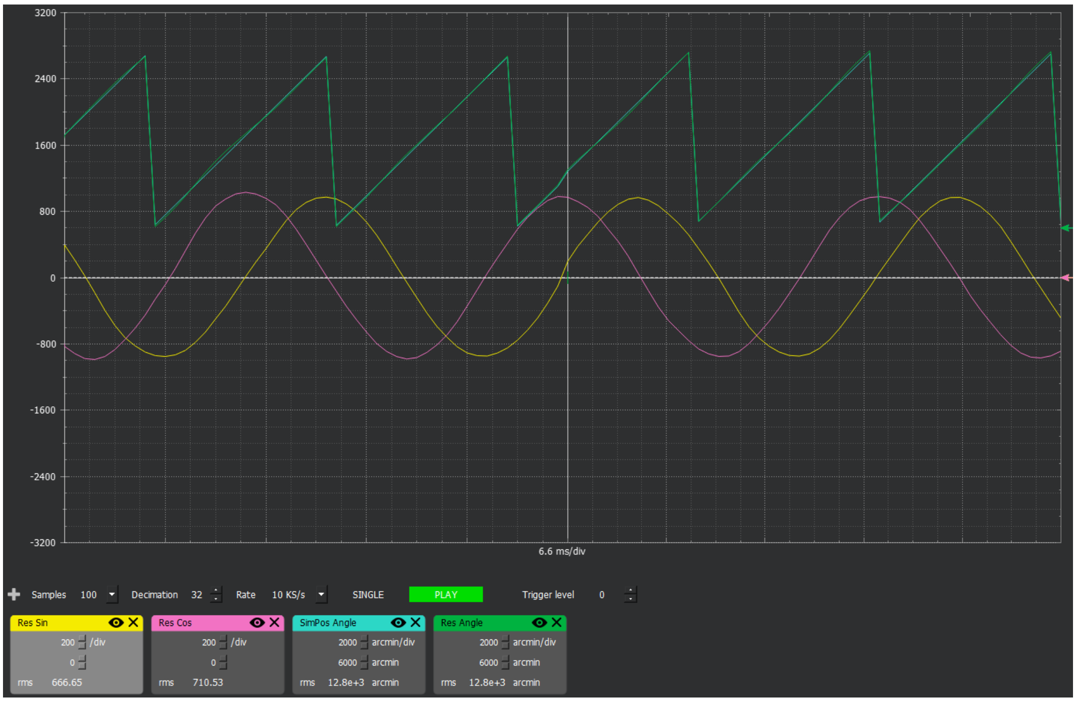
42	Setpoint D	5	‰/A
15	SimPos Speed	100	erpm

2. Once Cos/Sin are centered at zero and ranging from -1000 to 1000, you can set the poles of the machine and sensor, until the SimPos angle matches the resolver angle in frequency.

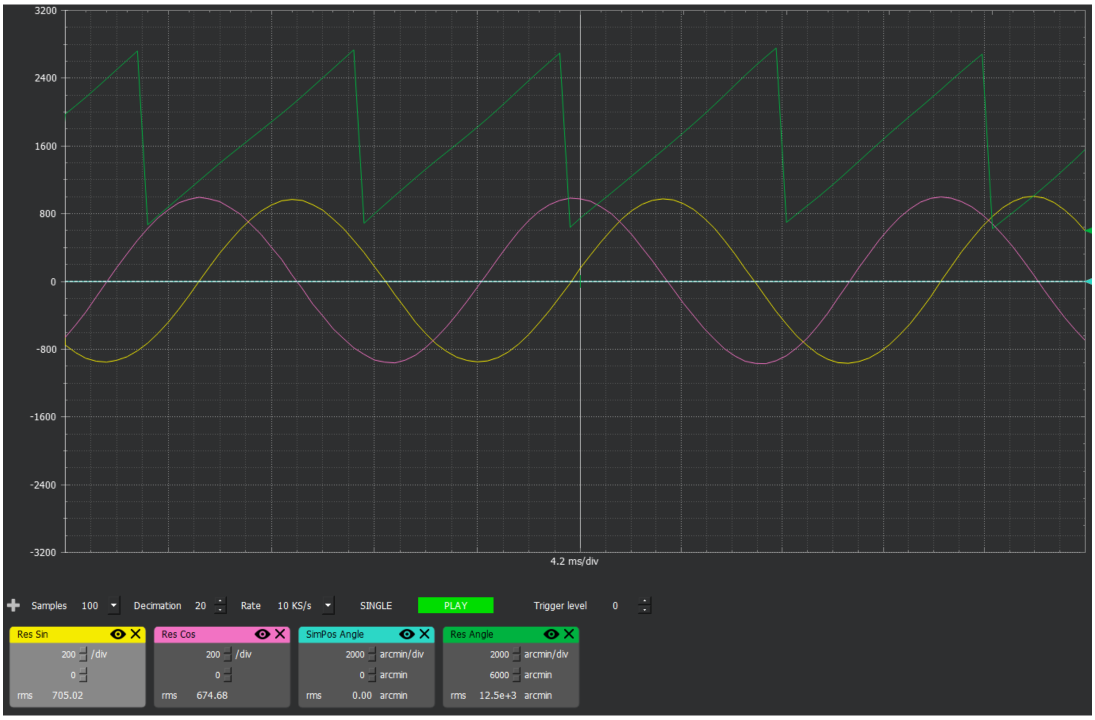
3. After that, it is necessary to match the D-axis offset. For that, alternate the SimPos from -100 to 100erpm and observe the phase delay between resolver and SimPos. Normally the resolver will drag some degrees (If the signals look small, increase the value of SimPos Speed to 1000).



The important thing is to match the phase both in forward and reverse. This is achieved increasing or decreasing the value of D-axis offset until match with the signal of Res angle.



4. If all is set and done, you should be able to change the position feedback source to resolver, and instead of Setpoint D, put a Setpoint on Q and machine should start spinning. Position feedback is calibrated.



11. ANNEXES

ANNEX 1: Default DBC (CAN1 ECU)

Message	Id	Direction	Period	Signal	Description	Uninit valu	Invalid valu	Length [bit	Units	Range
EMCU_STATUS_1	0x760	TX	10ms	Uptime	Control unit uptime in ms	0	N/A	32	ms	
				PwrStg_BitState	Power stage status bitfield	0	255	8	N/A	[DESAT, ASC, OCP, OVP, Enabled]
				EMCtrl_FOC_BitState	FOC controller status bitfield	0	255	8	N/A	[CurrImbalance, PosFBFault, ASC]
EMCU_STATUS_2	0x761	TX	10ms	App_State_App	Application state	0	255	8	N/A	1-127
				Current_D	Current feedback D-axis	0		16		
				Current_Q	Current feedback Q-axis	0		16		
				Volt_Modulus	Voltage modulus					
EMCU_STATUS_3	0x762	TX	10ms	Emachine_Speed	Machine speed	0		16		
				PwrStg_Temp	Power stage temperature	0		8		[-40,150]
				DP_Temp	Digital Platform temperature	0		8		[-40,150]
				Emachine_Temp_1	Machine temperature	0		8		[-40,150]
EMCU_STATUS_4	0x763	TX	100ms	Emachine_Temp_2	Machine temperature	0		8		[-40,150]
				Cmd_Src_App	Commands source	0	7	3	N/A	
				Ctrl_Type_App	Control type	0	7	3	N/A	1:Off,2:Scalar,3:Vectorial
				Ctrl_Mode_App	Control mode	0	15	4	N/A	1:DuCyABC,2:DuCyDQZ,3:Current,4:Torque,5:Speed,6:Position
EMCU_STATUS_5	0x764	TX	100ms	PosFb_Src_App	Position feedback source	0	15	4	N/A	1:SimPos,2:Sensorless,3:AbsEnc,4:IncEnc,5:Res,6:xMR,7:BiSS
EMCU_STATUS_6	0x765	TX	500ms	Dem_Fault				16		
EMCU_SETPOINT_1	0x660	RX	100ms	App_State_Req		0	255	8		[0 (Invalid), 1 (Off), 2(Ready), 3(On - Current mode), 4(On – Torque mode)]
				Current_D	Current setpoint	0		16		[-1000,1000]
EMCU_SETPOINT_2	0x661	RX	10ms	Current_Q	Current setpoint	0		16		[-1000,1000]
EMCU_SETPOINT_3	0x662	RX	10ms	Torque	Torque setpoint	N/A		16		[-1000,1000]

Revision					
Version	**Date**	**Description**	**Compiled**	**Verified**	**Approved**
V01	10/06/23	Creation of the document	A.Souto	J.Catalina	
V02					
V03	20/11/23	Features updated. Cooling removed	A.Saenz	A.Souto	
V04	01/12/23	Product warranty updated	A.Souto	A.Saenz	
V05	14/12/2023	HV Cable mounting render updated	A.Souto	A.Saenz	
V06	20/12/2023	Auxiliary cables updated	A.Saenz	A.Souto	